IBM FileNet P8 V5.6.x

Day 5

Storage & Security Administration

Storage Areas

Security Model

LDAP / VMM

Storage Configuration | FileNet Security Model | LDAP Integration (VMM)

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Learning Objectives – Day 5

Storage
Describe FileNet P8 storage architecture: areas, policies, content caching

Storage
Create and configure file system storage areas in ACCE for TrainingOS

Storage
Define and assign storage policies for automated document lifecycle management

Security
Explain Access Control Lists, security inheritance, and role-based access

Security
Configure object, class, and folder-level security using ACCE

LDAP
Integrate WebSphere VMM as the LDAP repository for FileNet P8

LDAP
Configure LTPA authentication and form-based login via WAS Admin Console

LDAP
Manage user/group resolution from LDAP into the FileNet security framework

01

Storage Configuration

Storage Areas · Storage Policies · Content Caching

Storage Components Overview

Storage Area

Logical grouping mapped to a physical device. Stores document binary content.

Storage Policy

Rules that route documents to specific storage areas based on class or age.

Content Cache

High-speed temporary store for frequently accessed content. Reduces retrieval latency.

Fixed Content Device (FCD)

WORM-compliant storage for regulated content retention (SEC, HIPAA, SOX).

Storage Area Types in FileNet P8 V5.6.x

File System Storage Area

Stores document binary content on a local or network file system (NFS or CIFS share).

Most common type used in development and training environments.

Training Lab Path:
C:\FileNetStore\TrainingOS\FS01

Affinity Cost: 10 (primary)
Max Size: No limit (configurable)
Accessibility: Online / Offline / Disabled

Fixed Content Device (FCD)

Write-Once-Read-Many (WORM) compliant storage for regulatory content retention.

Integrates with IBM Tivoli Storage Manager (TSM), EMC Centera, NetApp SnapLock, Hitachi HCP.

Required for compliance with:
- SEC Rule 17a-4
- HIPAA
- SOX

Retention period set at area level.
Documents cannot be deleted before expiry.

Advanced Storage Area

Policy-driven intelligent storage tiering with support for object store backends and S3-compatible cloud storage.

Ideal for hybrid on-premises and cloud deployments.

Supports automatic migration of aging content from fast SAN to low-cost cloud tiers.

Requires IBM Content Collector or FileNet Advanced Storage configuration.

Database Storage Area

Stores small document content (<10 MB recommended) as BLOBs directly inside the DB2 or Oracle object store database.

Advantages:
- No separate file system required
- Content backed up with database
- Simplified architecture for small deployments

Not recommended for large files or high-volume production.

Storage Policies & Content Caching

Storage Policy Flow

Document Created

New document saved in TrainingOS object store

Policy Evaluated

CE checks assigned storage policy for the document class

Initial Storage

Written to FS_TrainingStore_01 (Affinity Cost: 10)

90-Day Trigger

After 90 days, migration to FS_TrainingStore_02 (Cost: 50)

Content Cache Area

Location	Local SSD on CE/ICN server. NOT on storage area disk.
Eviction	LRU policy when max size reached. Configurable retention period.
Hit Ratio	Monitor in ACCE > Storage > Cache Stats. Target: >80%
Sizing	Set to 2x average daily retrieval volume for optimal performance
Cluster	Separate cache per node to avoid contention in WAS cluster

02

FileNet Security Model

Access Control Lists · Security Inheritance · Role-Based Access

FileNet P8 Security Architecture

Layer 5: Transport Security

SSL/TLS HTTPS for all CE/ICN API calls. Port 9443 (WAS HTTPS).

Layer 4: Application Security

WAS LTPA tokens, J2EE security roles, form-based authentication.

Layer 3: Content Engine ACLs

Object-level ACLs enforced by CE at every API call. Allow/Deny per principal.

Layer 2: Database Security

DB2 schema permissions for CEADMIN. Object store data is protected at DB level.

Layer 1: OS File System

File system permissions on storage area directories (NTFS ACLs on Windows).

Enforcement is bottom-up: OS → Database → CE → Application → Transport

Access Control Lists (ACLs)

ACL Structure

Object (Document / Folder)

ACE 1	**ceadmin**	
Allow	Full Control	
ACE 2	**#AUTHENTICATED-USERS**	
Allow	View + Read	
ACE 3	**filenet_managers**	
Allow	Read + Write	
ACE 4	**contractor_group**	
Deny	Write / Delete	

Permission Levels

View Content
Read document binary/rendition

Read
Read metadata and properties

Write
Modify metadata and properties

Major Version
Check in a major version

Minor Version
Check in a minor version

Create Child
Create objects in a container

Delete
Remove an object permanently

Change Security
Modify an object's ACL (most powerful)

Security Inheritance & Role-Based Access

Inheritance Hierarchy

Object Store (TrainingOS)

Top-level ACL — all objects inherit unless overridden

Folder (Root)

Inherits from Object Store + own ACEs

Subfolder (HR_Confidential)

Can break inheritance — isolated ACL

Document Class (Document)

Class ACL applies to all instances

Document Instance

Inherits from folder + class. Can add ACEs.

Recommended RBAC Roles

FileNet_Administrators

Full Control — all objects, classes, security

FileNet_ContentManagers

Create / Read / Update / Delete documents

FileNet_Reviewers

Read-only: view content and metadata

FileNet_RecordsManagers

Manage retention, legal hold, disposition

FileNet_DevTeam

Create classes and property templates

FileNet_ReportingUsers

Read-only metadata for analytics/reports

03

LDAP Integration

WebSphere Virtual Member Manager (VMM) · Authentication · Group Management

WebSphere VMM – Authentication Architecture

1

Browser / ICN Client

User opens ACCE or ICN. HTTP request to WAS.

2

LTPA Token Check

WAS checks for valid LTPA cookie. SSO if valid.

3

WAS Login Page

No token: redirect to form-based login page.

4

VMM → LDAP Bind

WAS VMM binds to LDAP server to verify credentials.

5

LTPA Token Issued

On success, WAS issues encrypted LTPA token (30 min).

6

CE ACL Evaluation

CE extracts username, gets group memberships from VMM, evaluates ACL.

Configuring LDAP in WebSphere VMM

WAS Admin Console Navigation Path

```
Security › Global Security › Federated repositories › Manage Repositories › Add LDAP repository
```

Training Lab Configuration Values

Repository ID	`TrainingLDAP`
Primary Host	`localhost`
Port	`389`
Bind DN	`uid=wpsbind,o=defaultWIMFileBasedRealm`
Base Entry DN	`o=defaultWIMFileBasedRealm`
Realm	`defaultWIMFileBasedRealm`

VMM Key Concepts

Federated Repository	Unifies multiple identity stores into one namespace for WAS
Realm	Security domain — defaultWIMFileBasedRealm in training lab
Bind DN	Service account VMM uses to search the LDAP directory
Group Membership	Resolved at runtime — CE queries VMM on every ACL check

LTPA Authentication & Group Management

LTPA Token-Based SSO

Format	Encrypted base64 cookie. Contains username, expiry, digital signature.
Timeout	Default 30 min. Change in WAS: Security > Global Security > LTPA.
SSO Domain	Set .training.ibm.com for SSO across multiple WAS apps on same domain.
Key Sync	All WAS cluster nodes must share the same LTPA key file (export/import).
CE Validation	CE decrypts token with shared LTPA key on every API call. No DB lookup.

LDAP Group Lookup Flow

1. User API Request

Content Engine receives the request

2. Extract Username

Username decoded from LTPA token

3. VMM Group Query

CE calls VMM.getMembership(username)

4. LDAP Recursive Search

VMM resolves nested group memberships

5. ACL Evaluation

CE checks object ACL against user + groups

Day 5 Lab Exercises

LAB-05-01 · 45 min
Configure Storage Areas

- Create C:\FileNetStore\TrainingOS\FS01 directory
- Create FS_TrainingStore_01 in ACCE (Affinity: 10)
- Set as default storage area for TrainingOS
- Upload test doc and verify physical storage

LAB-05-02 · 30 min
Configure Storage Policies

- Create STD_Policy_TrainingOS (90-day migration)
- Initial: FS_TrainingStore_01 → Dest: FS_TrainingStore_02
- Assign as object store default policy
- Verify new docs use the correct storage area

LAB-05-03 · 40 min
Implement Document Security

- Review Root Folder default ACL in ACCE
- Create HR_Confidential folder with restricted ACL
- Break inheritance and set explicit ACEs
- Configure Document class-level security

LAB-05-04 · 35 min
Configure LDAP via VMM

- Verify VMM config in WAS Admin Console
- Review LDAP repository connection settings
- Test user/group lookup via Manage Users
- Configure LTPA timeout and verify settings

Day 5 Complete

Storage & Security Administration

Storage Areas & Policies

File system, FCD, advanced storage, policy-driven tiering

Security Model & ACLs

Object ACLs, inheritance, RBAC with LDAP groups

LDAP Integration (VMM)

WebSphere VMM, LTPA SSO, group resolution

Next: Day 6 – Records Management & Retention

IBM FileNet P8 V5.6.x | Rajesh Pasham